

Vedansh Somani

Undergraduate Student

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Profile

Robotics & AI undergraduate specializing in machine learning, reinforcement learning, and autonomous systems. Experienced in developing deep learning models, RL agents, and ROS2-based robotic systems, with additional experience building full-stack applications.

Education

BE - Robotics and AI | CGPA: 9.11 09/2023 – Present
Dayananda Sagar College of Engineering Bengaluru, India

Professional Experience

STEM Trainer 04/2024 – 04/2025
Beylabee Bengaluru, India

- Delivered hands-on STEM and programming workshops, guiding students through robotics, electronics, and software-based projects.
- Designed and explained technical concepts through practical demonstrations, improving student understanding of engineering and computational principles.

Skills

Machine Learning and Reinforcement Learning

- PyTorch
- PPO
- DQN (Deep Q-Learning)
- Stable-Baselines3
- Neural Networks
- Deep Learning
- Reward Shaping
- Curriculum Learning

Robotics and Autonomous Systems

- ROS2
- Nav2
- SLAM
- Motion Planning
- Robot Control
- MuJoCo

Programming Languages

- Python
- C++
- HTML/CSS
- JavaScript
- Nodejs
- React
- Flutter

Embedded Systems and Tools

- ESP32
- Arduino
- Microcontrollers
- Git
- Linux
- TensorBoard

Projects

STRIDE v1 - RL Based Quadruped Locomotion (Unitree Go2)

06/2026 – Present

Python, MuJoCo, PPO, PyTorch

- Developed a PPO-based locomotion framework for a 15 kg Unitree Go2 quadruped in MuJoCo, training a single neural policy from scratch for gait generation and disturbance recovery using PMTG-based gait generation, residual joint control, and PD tracking.
- Evaluated robustness against impulsive mid-gait disturbances across 8 directions, achieving 100% recovery at 90 N (~0.6× body weight) on flat terrain and 100%/91.7% recovery at 75/90 N on procedurally generated uneven terrain (0.15 m elevation variation).

Autonomous Mobile Robot

02/2026 – 05/2026

ROS2, Nav2, SLAM, LiDAR

- Developed an autonomous mobile robot using ROS2, LiDAR-based SLAM, and Nav2 for localization and navigation.
- Integrated SLAM, localization, path planning, and obstacle avoidance pipelines on a differential-drive platform.

Chain Reaction + RL

05/2026 – 05/2026

Python, PyTorch, Deep Q-Learning

- Recreated the Chain Reaction board game in Python/Pygame and developed a DQN agent with CNN-based state representation using PyTorch.
- Built an RL training pipeline with curriculum learning, reward shaping, and heuristic-guided action filtering, achieving 57% win rate against a greedy baseline.

4WD Rover

07/2025 – 08/2025

ESP32, Embedded Systems

- Built a wireless 4WD rover using ESP32 and Cytron motor control with multiple input modes including Bluetooth, web interface, and RF controller.
- Designed embedded control logic for real-time motor actuation and remote operation.

Chat Application

12/2025 – 02/2026

Flutter, Node.js, Socket Programming

Developed a full-stack mobile chat application with real-time socket communication, backend services, database integration, and encrypted authentication.

Courses

Neural Networks - Zero to Hero

06/2026 – Present

Andrej Karpathy

Bengaluru

Implemented neural networks from scratch, covering backpropagation, optimization, MLPs, CNNs, and transformer architectures using PyTorch.

Data Analysis with Python

04/2026 – 04/2026

IBM Developer Skills Network

Bengaluru

Covered data preprocessing, exploratory data analysis, statistical analysis, and visualization using Python libraries including NumPy, Pandas, and Matplotlib.